Name: _____ Date: _____

Unit Vectors — What are the hats in $\vec{v}(t) = v_{0x}\,\hat{\imath} + (-9.8t)\,\hat{\jmath}$ doing?

SYNOPSIS

You have already written one. In Note 3 the launched ball's velocity came out as $\vec{v}(t) = v_{0x}\,\hat{\imath} + (-9.8t)\,\hat{\jmath}$, and those two hatted symbols did real work without ever being defined. This note names them. A unit vector is **pure direction**: magnitude exactly 1, and no units at all. Everything with a size and a unit sits in front of it. Once you have $\hat{\imath}$ and $\hat{\jmath}$, every vector in the plane is two ordinary numbers, and adding vectors becomes collecting like terms.

PERFORMANCE EXPECTATIONS

1. **State what makes a vector a unit vector** — magnitude exactly 1, no units — and explain why the hat contributes nothing to the units of the quantity it describes.
2. **Write a vector in unit-vector form** from its components, and read a unit-vector expression back as a magnitude and a direction.
3. **Multiply a vector by a scalar** and say what changes, what cannot change, and why a negative scalar is a statement about direction rather than a negative magnitude.
4. **Add two vectors given in unit-vector form** by collecting the $\hat{\imath}$ and $\hat{\jmath}$ terms separately, then recover the magnitude and direction of the result.

Instructions and Questions

1. **Predict first.** In $\vec{v}(t) = v_{0x}\,\hat{\imath} + (-9.8t)\,\hat{\jmath}$, what does $\hat{\jmath}$ contribute to the *units* of $\vec{v}$? Circle one, then say why.

   m/s m s Nothing — it has no units

   Why?

2. **Read it in pieces.** Fill in the table for $\vec{v}(t) = v_{0x}\,\hat{\imath} + (-9.8t)\,\hat{\jmath}$.

Piece	Has a size?	Units	What it tells you
v_{0x}			
$\hat{\imath}$			
$-9.8t$			
$\hat{\jmath}$			

Why is the minus sign in $-9.8t$ there?

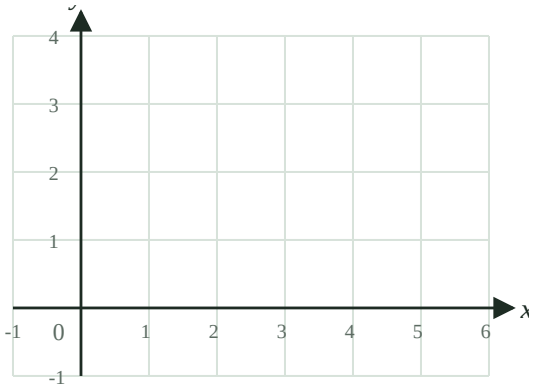
3. **The definition — three things.** A unit vector has

(a) magnitude: $|\hat{\imath}| = |\hat{\jmath}| =$ _____ (b) direction: $\hat{\imath}$ along _____, $\hat{\jmath}$ along _____ (c) units: _____

Do $\hat{\imath}$ and $\hat{\jmath}$ rotate when the object turns? _____

4. **Magnitude is never negative.** The vector $-4\,\hat{\imath}$ has magnitude _____ and points along _____. A classmate says "the magnitude of $-4\,\hat{\imath}$ is -4." What is wrong?

5. **Build a vector on the grid.** One grid square = 1 m. Draw $\hat{\imath}$ and $\hat{\jmath}$ at the origin (one square each). Then build $\vec{V} = 4\text{ m}\,\hat{\imath} + 3\text{ m}\,\hat{\jmath}$: four metres' worth of $\hat{\imath}$, then three metres' worth of $\hat{\jmath}$. Draw $\vec{V}$ from the origin to the end.



$|\vec{V}|$ = _____ $\theta = \tan^{-1}($ _____ $) =$ _____ above +x

6. **What a number in front can — and cannot — do.** Let $\vec{A} = (2\text{ m})\,\hat{\imath} + (1\text{ m})\,\hat{\jmath}$.

   $3\vec{A} =$ _____ $\hat{\imath} +$ _____ $\hat{\jmath}$ $\quad -\vec{A} =$ _____ $\hat{\imath} +$ _____ $\hat{\jmath}$

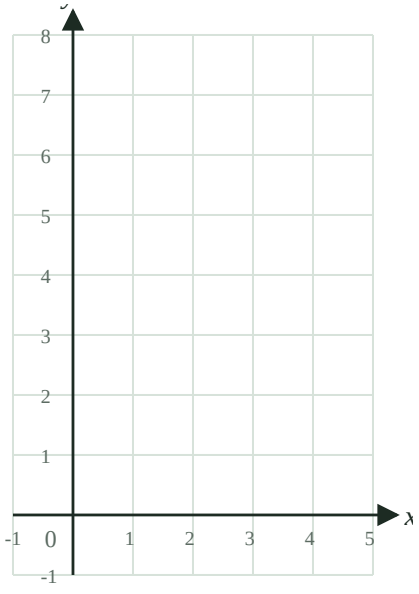
   A scalar can change:

   A scalar can *never*:

7. **Add by collecting like terms.** $\vec{A} = (3.0\text{ m})\,\hat{\imath} + (5.0\text{ m})\,\hat{\jmath}$ and $\vec{B} = (-1.0\text{ m})\,\hat{\imath} + (2.0\text{ m})\,\hat{\jmath}$.

   $$\vec{A} + \vec{B} = (\ _____\ )\text{ m}\,\hat{\imath} + (\ _____\ )\text{ m}\,\hat{\jmath} = _____\ \hat{\imath} + _____\ \hat{\jmath}$$

   Draw $\vec{A}$, then $\vec{B}$ head to tail, then $\vec{A} + \vec{B}$ from the origin.



   $|\vec{A} + \vec{B}| =$ _____ $\quad \theta =$ _____

8. **Where this holds.** Could you add a displacement (in m) to a velocity (in m/s) this way? Why or why not?

9. **Watch the quadrant.** $\vec{C} = (-2.0\text{ m})\,\hat{\imath} + (-7.0\text{ m})\,\hat{\jmath}$. Your calculator gives $\tan^{-1}(-7.0 / -2.0) =$ _____.

   Sketch $\vec{C}$. Which quadrant is it really in? _____ Direction: _____

10. **Two notations, same three arrows.** $\hat{\imath}$ = _____, $\hat{\jmath}$ = _____, $\hat{k}$ = _____. Pick one and stay with it inside a single problem.

> **Looking ahead — Note 4.** The launcher tilts: $\vec{v}(t) = v_0\cos\theta\,\hat{\imath} + (v_0\sin\theta - 9.8t)\,\hat{\jmath}$.
>
> Which term changes with time? _____ Which stays constant? _____

Exit Ticket

1. The magnitude of $\hat{\imath}$ is…

   1 metre | Exactly 1, with no units | Whatever one grid square represents | Undefined until you choose axes

2. "The magnitude of $-4\,\hat{\imath}$ is -4." What is wrong?

   Nothing — it is correct | Magnitude is 4; the minus sign gives the direction, −x | Magnitude is 4 and it points along +x | You can't multiply a unit vector by a negative number

3. $\vec{A} = (6.0\text{ m})\,\hat{\imath} + (2.0\text{ m})\,\hat{\jmath}$ and $\vec{B} = (-2.0\text{ m})\,\hat{\imath} + (5.0\text{ m})\,\hat{\jmath}$. The $\hat{\jmath}$ component of $\vec{A} + \vec{B}$ is _____ m.

4. Multiplying a vector by the scalar 3 will…

   Rotate it by 3° | Make it 3× as long, same direction | Make it 3× as long and reverse it | Change its units

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